

Version #	Date	Details of change	Page / slide no.

Risk and issue management

Learning objective:

23. Understand risk and issue management as the ability to identify and monitor risks (threats and opportunities); plan responses to those risks and respond to issues.

23a) Understand the benefits of risk and issue management and the role of contingency planning in projects.



Risk management

Key aspects

- Risk management allows individual risk events and overall risk to be understood and managed proactively, optimising success by minimising threats and maximising opportunities.
- Risk management is influenced by the risk appetite of the investing organisation that will provide the tolerance levels on what is acceptable or not.
- The risk management process is iterative to reflect the dynamic nature of project work.

Benefits

- Supports more realistic plans, schedules and budgets.
- Increases the likelihood of schedules and budgets being kept to.
- Greater accuracy in allocated contingencies through more meaningful assessment.
- Supports continuous learning and improvement.
- All contributing to increasing stakeholder confidence.

Issue management

Key aspects

- The process used to manage issues is a simple concept focused on understanding the issue, in order to prioritise and assign owners for a timely resolution.
- Issue resolution requires support from the project sponsor, who may escalate to the governance board.
- The management of issues are tracked from identifying to resolving the issue and may require a re-plan of the baselined project management plan.

Benefit

- Provides the structure and escalation routes required to resolve issues in a timely manner.
- Raises visibility among senior stakeholders who have access to organisational resource to support resolution.
- Provides a consistent way to document issues so they can be assessed against the plan.
- Enables re-prioritisation and decision making.
- Ultimately it enables issues to be resolved efficiently so work can progress preventing further delays.
- Informs continuous learning and improvement.

Contingency planning

- Contingency is the resource set aside for responding to identified and unidentified risks, also known as the 'unknown unknowns'.
- Contingency is not hidden extra time or money to deliver planned scope: it is clearly identified and is there to cover the gap between the plan and known risks.
- In addition to the contingency for known risks, some projects will have access to a management reserve for unidentified risks or identified risks that have very low likelihood of materialising but if they did would have a very high impact.
- Contingency is most typically expressed in two ways: budget and time. It's for dealing with uncertain events that have an impact on either the project costs / financial benefits or the project timescales.
- However, if using an iterative life cycle and timeboxes, the project would consider contingency in terms of scope or quality.
- Where risks are identified, the planned contingency has provision to cover the gap, and the project manager requests to draw down some of the contingency to manage mitigation.



23b) Understand the purpose and importance of each stage in a risk management process (such as identification, analysis, monitoring, escalation, response and closure) and an issue management process (such as logging and analysis, escalation, and assignment of actions) and understand why these stages are different for linear and iterative life cycles.

Risk management process

Risk management must be initiated by first establishing the risk management plan for the project. This plan provides the roles and responsibilities, how risks are to be assessed and the impact evaluated, agreed tolerances for risk thresholds, reporting and escalation points. It also serves as an important reference for all stakeholders.

Initiation can be the first step; a typical risk management process will then include the following steps:

Identification

- Identifies the risk events relevant to a project considering risk appetite.
- Ongoing process throughout the project life cycle.
- There are several tools and techniques that can be used to draw out knowable risks.
- Once identified, risks should be documented with the appropriate information in the project risk register and an owner assigned to assess and manage.
- When working in an iterative life cycle, risk identification has a different focus, using a risk-based approach mindset to influence the design of each iteration, while continuing to look at what is risky, why and what can be done about it during deployment.

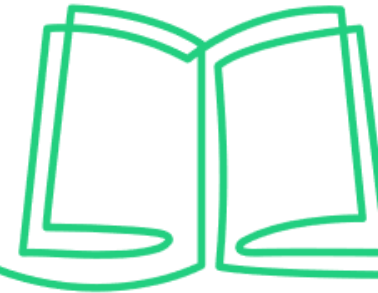
Analysis

- Assesses the relative impact of the identified risk on the project objectives.
- Undertaken by the project manager, with the support of the project team and subject matter experts, using qualitative and or quantitative techniques.
- Qualitative analysis looks at the impact on objectives, assessing the probability / likelihood of occurrence and the size of impact.
- Quantitative analysis determines the combined effect of risks on objectives and for focus on specific risk-based decisions.
- When working in an iterative life cycle, risks will be analysed for each iteration; the issue of each release could then prompt different risks to be analysed.

Monitoring and Escalation

- Monitoring of risks for occurrence and knowing when to escalate.
- If risk thresholds are reached and considered to have potentially critical consequences to the project objectives, escalation would be required.

Risk management process (continued)



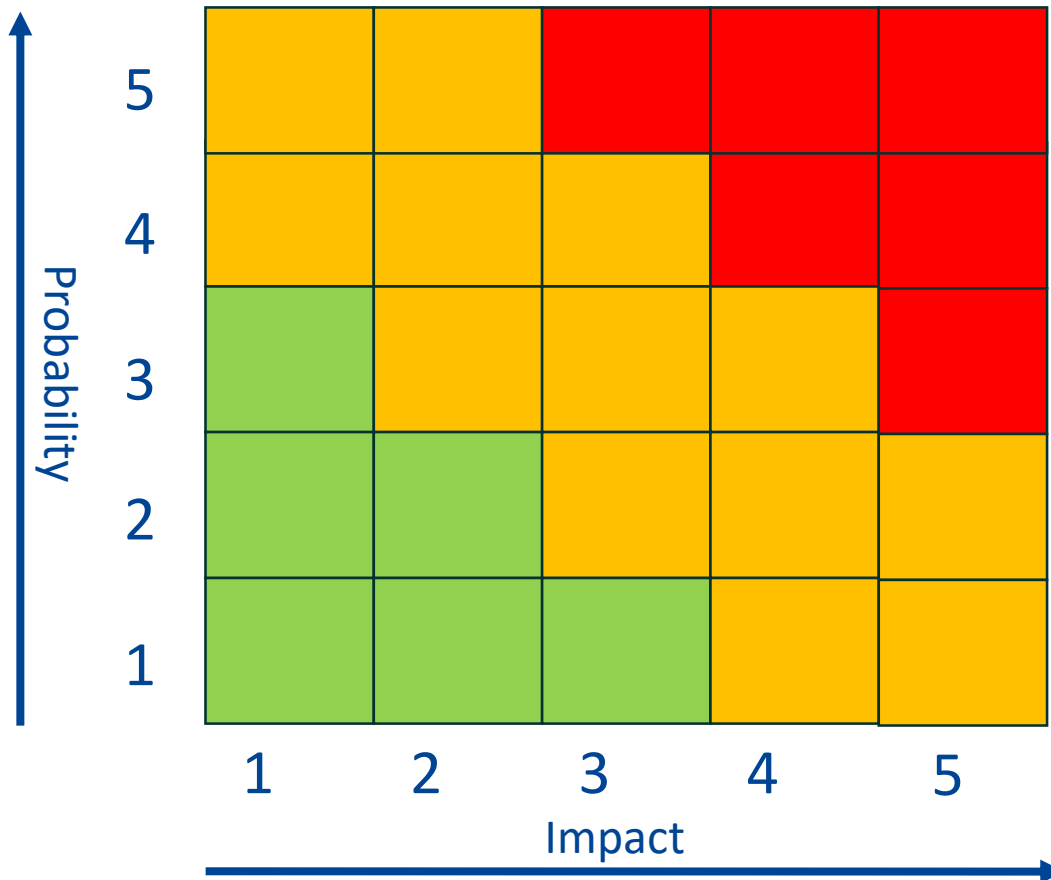
Response

- Considers the appropriate response to put in place to deal with the risk, whether proactive or reactive.
- Determines whether the risk is deemed a threat or an opportunity.
- Determines whether it makes sense to proactively invest in mitigation activity to bring the risk back within tolerance levels.
- Secondary risks or residual risks may emerge because of the response and should be considered.
- Focus is on putting a risk efficient solution in place, increasing the likelihood of project success.

Closure

- Closes the risk in the risk register when it no longer poses a risk to the project. This could be because it has been removed, been responded to, occurred or passed the point it was likely to occur.
- As the last step in the process, it is important to update the risk register with information relating to the risk closure and lessons learned to avoid unnecessary work of monitoring risks that will no longer occur and to support continuous improvement of project activities and processes.
- Any remaining risks at project closure should be communicated to stakeholders involved in the handover and adoption phase.

Probability/Impact Grid Example



Key:



Intolerable



Undesirable



Acceptable

Each identified risk is qualitatively assessed and placed on the grid, allowing the risks to be prioritised



Issue management process

Tracking the management of issues happens from identification through to resolution including any change control and replanning activities required to the baselined project management plan.

Log and analyse

- An issue register / log should be maintained and used by the project manager and team to record, analyse, prioritise and track the status until the issue is resolved.
- Prioritisation is based on the success criteria and benefits for the work, considering scope, quality, time, cost and benefits in the business case.
- Issue registers / logs are a source of knowledge for future projects.

Escalate

- The issue will need escalating to the project sponsor, who may then escalate to the governance board.
- The sponsor / governance board will help the project manager define the best course of action for a resolution.

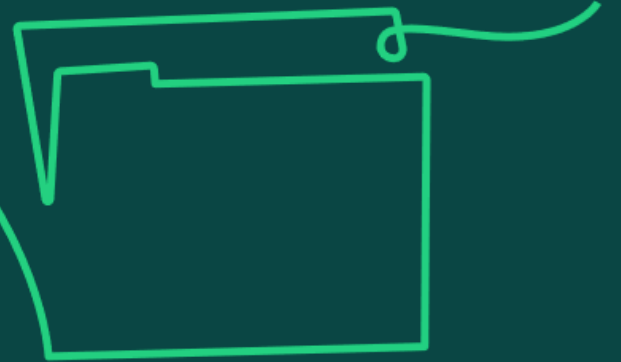
Assign actions

- Actions are assigned to a person or group best placed to resolve.
- Look to implement a resolution in a timely manner.

Apply change control

- Issues that result in a change to scope or the baselined plan will need to be taken through the change control process.
- This ensures all project outputs are updated and checked for no further impact to the project.

23c) Know proactive and reactive responses to risks (such as avoid, reduce, transfer or accept and exploit, enhance, share and reject).



Proactive and reactive responses to risks

Risks are categorised as threats or opportunities and there are two main types of responses to them: proactive or reactive.

PROACTIVE A planned and implemented response undertaken to address the likelihood of the risk occurring or the size of the impact if it did occur.	REACTIVE A provision for a course of action that will only be implemented if the risk materialises.
Threats • Avoid: Change objectives or practices to change the cause of the threat so it can no longer occur. • Reduce: Take action to reduce the probability of the threat occurring or the impact on the project. • Transfer: From one party to another, passing on the responsibility for bearing the impact of the threat.	Threats • Accept: No acceptable or viable approaches to take to avoid or reduce the threat. Any residual risk must be managed, and contingency plans implemented should the risk materialise.
Opportunities • Exploit: Change objectives or practices to implement a response to maximise the opportunity to gain maximum benefit / advantage. • Enhance: The probability of the opportunity eventually being exploited by increasing the likelihood and / or impact. • Share: Collaborating with other organisations to increase the chances of the opportunity being realised. • Reject: Decide not to take advantage of the situation as the opportunity is of little value or the work to gain the result is too great.	





23d) Understand why governance is important in risk and issue management.



Why governance is important

- Governance and risk management work together, providing systems of control for the project and giving confidence to stakeholders.
- Governance informs risk and issue management through processes, procedures, standardised methods, specialist expertise and resources.
- It sets out the arrangements for the organisation and project, the life cycle to be used for project delivery, the route for reporting and escalating, and provides assurance which is targeted where the greatest risks exist.
- Assurance is objective and independent, and governance relies on assurance to support effective decision making when it comes to managing risks and issues.
- Risk analysis provides guidance to assurance on where vulnerabilities lie and in return evidence from assurance improves risk assessment and informs decision making at decision gates.



Photo credit: Loic Leray